

**Time:** 6 minutes**Outcome:** Calculate midpoint price elasticity of demand and interpret the result**Difficulty:** ●●○ Intermediate

Price Elasticity of Demand



THE CORE IDEA

Price elasticity of demand (PED) is the absolute value of percentage quantity change divided by percentage price change. The midpoint method divides each change by the average of its starting and ending values, giving the same result in either direction. PED above one is elastic, one is unit elastic, and below one is inelastic.



HOW TO RECOGNIZE IT

- Use midpoint averages when comparing two price-quantity points.
- Close substitutes, time to adjust, a larger budget share, and a narrower market make demand more elastic.
- A straight demand curve has constant slope but varying elasticity.

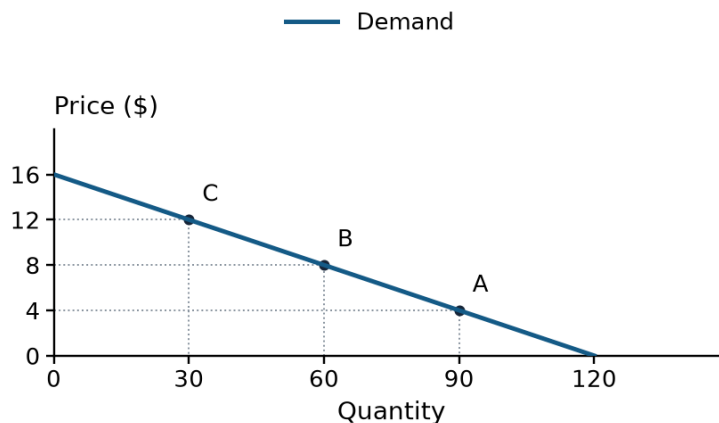


WATCH OUT

Elasticity compares percentage changes; slope compares changes in the original units. A steep-looking curve is not automatically less elastic.



WORKED EXAMPLE: MIDPOINT ELASTICITY



Between C and B, price falls from \$12 to \$8 while quantity demanded rises from 30 to 60. Using midpoint values, quantity changes by 66.7% and price changes by 40.0%, so $PED = 66.7\% / 40.0\% = 1.67$. Because 1.67 is greater than 1, demand is elastic over this interval: quantity responds proportionally more than price changes.



CHECK YOURSELF

Using points B (P = \$8, Q = 60) and A (P = \$4, Q = 90), what is midpoint PED, and how should demand be classified over that interval?

**READY?**

Return to the game and master this concept.